

Fine-tuning LLM for Counseling Application in Federated Learning Setting

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Abstract

LLMs hold significant potential for counseling applications, which can support the treatment of common mental health conditions. If LLMs can perform counseling tasks as effectively as human counselors, it could generate helpful, patient-specific advice to a wide range of patients, at any time, reducing financial and logistic barriers to mental healthcare. One way to train LLMs to perform counseling tasks like human counselors is to fine-tune it with real conversational data of patients and counselors, produced by audio-recording counseling sessions. Although that data would be highly valuable in incorporating human-like tone and fluency in LLM's responses, direct access to such data is often not feasible due to privacy concerns. Federated learning offers a solution by enabling collaborative model training across decentralized data sources without exposing the raw data. This study makes two main contributions. Firstly, we propose a novel application of fine-tuning LLM through federated learning : Counseling. Secondly, we conduct an experiment to show that federated learning does not limit the effectiveness of fine-tuning any more than centralized training for the novel application of counseling. Our experimental findings demonstrate that federated learning can fine-tune LLMs for counseling without compromising model performance, providing a privacy-preserving alternative to centralized approaches.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools; • **Computing methodologies** → Natural language generation.

Keywords

Federated Learning, LLM, mental health, counseling, chatbot, fine-tuning

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1 Introduction

More than 5% of the world population suffer from depression, which is the most common mental health condition globally [8]. One of the treatments to such mental health conditions is counseling. Counselors have conversations with depression patients and provide helpful advice and accurate information based on their personal context. However, according to a study investigating student mental health [5], most students have not sought help in times of crisis. Barriers preventing students from seeking help are stigma surrounding topics of mental health, low availability of human counselors, and inadequately trained counselors.

As a solution to this problem, this paper suggests fine-tuning LLM to perform counseling tasks like human counselors. Counseling LLM would be able to provide patient-specific advice to a wide range of patients, at any time, reducing financial and logistic barriers to mental healthcare. This can be done by using transcripts of real counseling sessions to incorporate human-like tone and fluency in LLM's responses.

However, use of transcripts from counseling sessions can be seen as an invasion of privacy. In fact, many patients are willing to share health data to improve services, but are still concerned with privacy issues [12]. One way to include such private counseling data in model training while protecting its privacy is federated learning.

Federated learning consists of a server and multiple clients, each having local data. The server sends training instruction and initialized model to each client. Each client trains the given model using their respective local data. Only the parameters of local training are sent back to the server. These parameters are aggregated on the server and the aggregate model is updated. This is repeated over multiple rounds. Only the parameters and not the local data are sent to the server, thus the privacy is protected [6].

From here on, the term "federated" would describe a setting where data is distributed to multiple clients. In contrast, the term "centralized" would describe a setting where data is collected in one place (on the server).

The main contributions of this paper are as follows:

- We propose a novel application of fine-tuning LLM through federated learning : Counseling.

- We conduct experiment to show that federated learning does not limit the effectiveness of fine-tuning any more than centralized training for the novel application of counseling.

2 Related Work

Recent studies have explored the use of large language models (LLMs) in specialized domains and under privacy-preserving settings. However, none have focused specifically on applying LLMs to counseling in a federated learning framework using private text data. This section reviews the most relevant works and outlines how our approach differs from them.

One of the relevant studies [14] fine-tunes pre-trained LLM in federated setting in order to answer math and medical questions. A key contribution of this study is its handling of unstructured data. Since not all clients have structured data (i.e. question and answer pair), the authors convert unstructured local data into structured data before training. While this study addresses the task of fine-tuning LLM through federated learning, it only focuses on math and medical domains and not counseling application.

Another study [15] also fine-tunes pre-trained LLM in federated setting in order to answer math and medical questions. A key contribution of this study is the use of data quality filter in order to conduct model training with high quality data only. While it again explores the task of fine-tuning LLM in federated setting, the application of the system for counseling is not discussed.

Some studies do explore applications of LLM in mental healthcare. Mental-LLM [13] perform the task of mental health prediction via online text data. Although this work is relevant to the broader domain of mental healthcare, it differs from our study in two key aspects: (1) it focuses on prediction rather than counseling or response generation, (2) models are trained in centralized setting, disregarding privacy protection.

While various studies explore either the task of fine-tuning LLM through federated learning or the application of LLM in mental healthcare, no current work applies fine-tuning LLM through federated learning specifically for the application of counseling.

Since counseling is a novel application of LLM fine-tuning through federated learning, experiments need to be conducted to ensure that the federated setting does not hinder the effectiveness of fine-tuning any more than the centralized setting for this specific application.

3 Proposed method

This study proposes an experiment to investigate whether or not federated setting limits the effectiveness of fine-tuning of LLM specifically for counseling application.

Pre-trained models will be compressed into PEFT models, which are then trained using dataset of transcripts between human patient and counselor. Identical training flow will be conducted in centralized and federated setting respectively and the results will be compared.

If the results are similar in federated and centralized, it can be concluded that the federated setting does not limit the effectiveness of fine-tuning LLM for counseling application. From this, we would conclude that federated learning can be a privacy-preserving alternative to centralized fine-tuning.

3.1 PEFT and LORA explained

PEFT stands for Parameter Efficient Fine Tuning. It is a method of efficiently fine-tuning an LLM by limiting the trainable parameters to a small portion of the whole model. LoRA is a type of PEFT that does not directly update the Weight Matrix = W of the original model, but uses low-rank matrices A and B to express the updated model W' as $W + AB$. It is lightweight, preserves the original model, and is more consistent than other PEFT methods[3, 4, 9], thus LoRA was selected for this experiment.

4 Experiment

4.1 Experiment Overview

We conducted our experiments using the Amod Mental Health Counseling Conversations dataset¹ to fine-tune pre-trained language models for counseling applications. Two data distribution schemes, federated and centralized, were compared to assess their impact on model performance. As backbone architectures, we employed Llama-3.2-3B-Instruct and Phi-3-mini-4k-Instruct, and evaluated the quality of generated responses using both GPT-Scoring and BERTScore.

Figure 1 shows the experiment overview. The general flow of the experiment is explained in Section 3. The right side of the figure shows centralized training and the left side shows federated training.

4.2 Rationale for pre-trained model selection

We selected two pre-trained models that are widely used in NLP community. Phi-3-mini-4k-Instruct model [7] produced a high score for MMLU benchmark with a relatively low number of parameters(3B). Llama-3.2-3B-Instruct model [2, 10] produced a high score for AGIEval benchmark with 3B parameters. These models were chosen for this experiment because both of them produce a high score for aggregate benchmarks with relatively light-weight parameters.

4.3 Rationale for dataset selection

Amod/mental_health_counseling_conversations was used as training data. This dataset contains 3.51k question and answer pairs between human patients and qualified counselor across various mental health topics. Due to the relatively small size of the dataset, except for 50 entries, all of the dataset was used for training. The excluded 50 entries were used for evaluation.

4.4 Experiment configuration

We fine-tune the meta-llama/Llama-3.2-3B-Instruct and Phi-3-mini-4k-instruct model on the Amod/ mental_health_counseling_conversations dataset using a federated learning setup facilitated by the Flower framework.

To reduce the memory usage and support training on consumer-grade GPUs, we use 4-bit quantization. Gradient checkpointing is also enabled to save memory during backpropagation.

In this configuration, the LoRA rank is set to 16, and the scaling factor is set to 64, resulting in an effective adaptation strength of 4.0. These values provide a good trade-off between expressiveness and

¹https://huggingface.co/datasets/Amod/mental_health_counseling_conversations

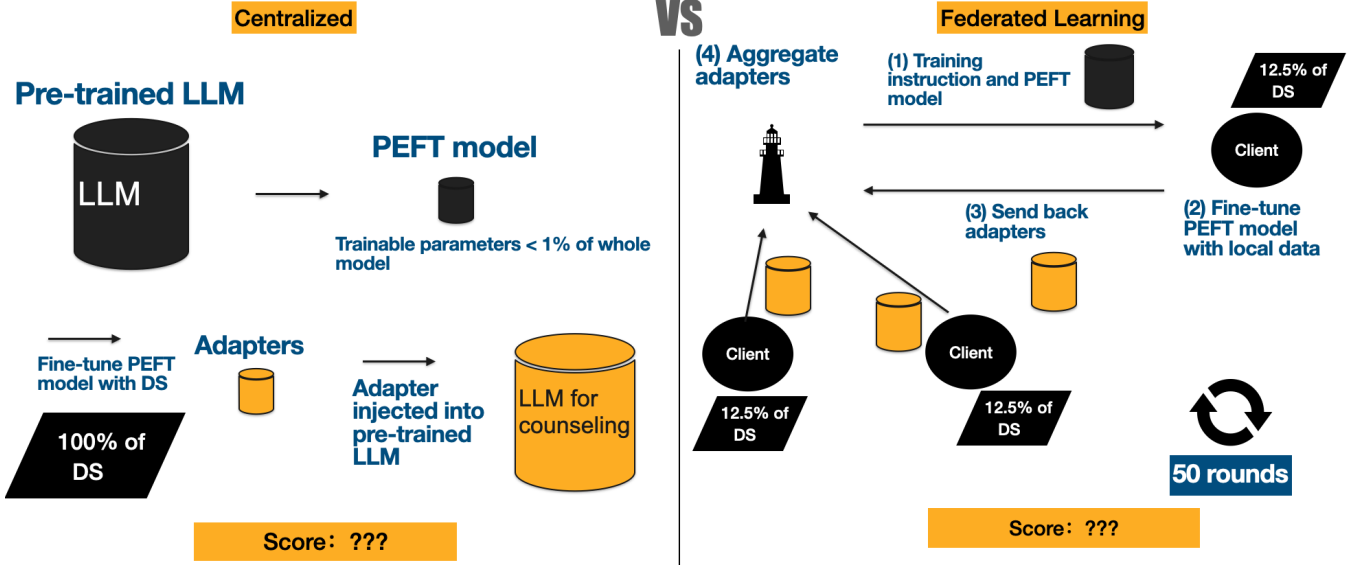


Figure 1: Experiment Overview : Pre-trained models will be compressed into PEFT models, which are then trained using dataset of transcripts between human patient and counselor. Identical training flow will be conducted in centralized and federated setting respectively and the results will be compared.

efficiency. LoRA is applied to all linear layers, allowing widespread adaptation across the model without modifying its core architecture.

The input sequence length is set to 512 tokens, which strikes a balance between capturing conversational context and maintaining computational efficiency. The learning rate is configured to vary between 5×10^{-5} and 1×10^{-6} to accommodate potential scheduling strategies.

The training process is performed over 50 federated rounds, with 8 clients participating in total and 50% selected randomly for each round. Each client performs a limited number of training steps (10 per round) with a small batch size and 3 epochs per local training phase, reducing the risk of local overfitting.

5 Evaluation

Once the model was trained in both centralized and federated settings, 50 entries from the dataset was used to evaluate the model. The mean and standard deviation for each of the following evaluation metrics were recorded.

5.1 BERTScore

BERTScore calculates the Cosine Similarity of the embedding of two sentences using the pre-trained model Sentence-BERT [11]. It provides a standardized, model-agnostic measure of semantic similarities between two sentences[1]. The formula is as follows.

$$\text{Cosine Similarity} = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

$$\text{Cosine Distance} = 1 - \text{Cosine Similarity}$$

The similarity between the responses of LLM and human counselors was calculated. The score of 1 indicates the highest similarity, while the score of 0 indicates no similarity.

5.2 GPTScoring

GPT-scoring was also used for evaluation as it can rate LLM's response based on user-engineered prompt, reflecting flexible criteria [15].

For GPT-scoring, two prompts were prepared, corresponding to two distinct evaluation criteria. The first criterion is similarity between LLM's response and human counselor's response to a given statement by patients. This criterion assumes that there is a ground truth; that is, the human counselor's answer is the only absolutely correct answer to the patient's statement. Based on this assumption, LLM's response received a high score if it was similar to that of human counselors. The following prompt was used:

Instruction: Please rate how similar the LLM's response is to the human counselor's response based on the input.

Patient's statement: [Input]

Human response: [human response]

LLM response: [LLM response]

Score:

The second criterion is helpfulness of LLM's response based on patients' statements. This criterion assumes that there is no ground truth; that is, there may be more than one suitable response to the patient's statement. Based on this assumption, not the similarity but the helpfulness of LLM's response to the patient's statement was measured. The following prompt was used:

Instruction: Please evaluate whether the LLM's answer is suitable and helpful for the patient's statement.

Patient's statement: [Input]

LLM response: [LLM response]

Score:

Both scores were out of 5, 0 indicating the lowest and 5 indicating the highest.

6 Results

6.1 Comparing performance between Federated and Centralized

Table 1 summarizes the mean scores of pre-trained models for each evaluation metric after centralized and federated fine-tuning. For each evaluation metric, p-value was calculated, representing the probability that the observed difference between centralized and federated could occur by chance.

Except for statistically significant(*), all p-values were above $\alpha=0.05$ with the smallest being 0.069. This means that, for most of the evaluation metrics, scores produced by centralized and federated were not significantly different. It must be noted that no significant difference between centralized and federated is desirable as it demonstrates that federated learning does not limit the effectiveness of fine-tuning for counseling any more than centralized. Therefore, the results show that federated learning can be a privacy-preserving alternative to centralized learning.

Table 1: summarizes the scores of each pre-trained model after centralized and federated fine-tuning. For each evaluation metric, p-value was calculated, representing the probability that the observed difference between centralized and federated could occur by chance.

	GPT scoring (0–5) min=0, max=5		BERT score (0–1) min=0, max=1
	Similarity	Helpfulness	
Central Llama	2.36	2.80	0.6353
Federated Llama	2.34	2.76	0.6449
p-value	0.8693	0.7613	0.069
Central Phi	2.08	2.44	0.6376
Federated Phi	2.04	2.62	0.6226
p-value	0.7308	0.1266	0.0026*

6.2 Comparing performance between based model and fine-tuned models

Table 2 summarizes the mean scores of base models and fine-tuned models for each evaluation metric. For each evaluation metric, p-value was calculated, representing the probability that the observed difference between base model and fine-tuned model could occur by chance. The plus and minus sign in front of the p-value represents if the score has increased or decreased after fine-tuning.

*(The scores for fine-tuned Llama and Phi are those of centrally fine-tuned Llama and centrally fine-tuned Phi)

First of all, Llama’s score for GPT-scoring (similarity) improved significantly with p-value of less than 0.0001. Llama’s BERTScore and Phi’s BERTScore also improved but the p-values are both greater than $\alpha = 0.05$, indicating that the improvement is not

significant. Llama’s GPT-scoring (helpfulness), Phi’s GPT-scoring (similarity and helpfulness) all decreased after fine-tuning, suggesting areas for improvement. One change that could improve the performance of fine-tuning would be to use learning rate scheduler to adjust learning rate flexibly and avoid over-fitting problem.

Table 2: summarizes the mean scores of base models and fine-tuned models for each evaluation metric. For each evaluation metric, p-value was calculated, representing the probability that the observed difference between base model and fine-tuned model could occur by chance. The plus and minus sign in front of the p-value represents if the score has increased or decreased after fine-tuning.

	GPT scoring (0–5) min=0, max=5		BERT score (0–1) min=0, max=1
	Similarity	Helpfulness	
Base Llama	1.78	3.04	0.63528
Fine-tuned Llama	2.36	2.8	0.64515
p-value	(+) <0.0001	(-) 0.0668	(+) 0.0572
Base Phi	2.56	3.4	0.63613
Fine-tuned Phi	2.08	2.44	0.63758
p-value	(-) 0.0001	(-) <0.0001	(+) 0.7243

7 Conclusion

Our experimental results highlight a novel insight that federated learning can facilitate fine-tuning of LLM for counseling as effectively as centralized training. This means that federated learning can be a privacy-preserving alternative to centralized learning in fine-tuning LLM for counseling application. In addition, fine-tuned Llama model performed significantly better than the base model for GPT-scoring (similarity). Fine-tuning did not improve the score significantly for other evaluation metrics and for Phi model, suggesting areas for improvement, such as suboptimal learning rate and other tuning parameters. Further experiments with revised experiment settings would clarify how much fine-tuning is able to improve LLM’s counseling ability. Lastly, it might also be meaningful to explore other methods of improving LLM’s counseling ability besides fine-tuning. Since the domain of counseling does not have clear ground truth and human counselor’s response in the training data is not the only suitable response, fine-tuning alone might not be sufficient to develop a helpful counseling LLM.

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